

Disrupt Quant 2026

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1. Hypothesis

Within each sector, I expect the highest-carry asset to outperform the lowest-carry asset over the next session, from open to close. Carry may reflect compensation for holding an asset, but the data dictionary defines it only as a slowly varying characteristic. The economic explanation remains a hypothesis.

2. Evidence

My development analysis ranked signals against next-session open-to-close returns using daily Spearman information coefficients (ICs), both across 24 assets and within four-asset sectors. The research draft reported carry mean ICs of 0.024 across assets and 0.030 within sectors, positive ICs in each development year, and rank persistence of 0.99.

Conflicting evidence included a 1-day return IC of -0.047 with rank persistence of 0 and a sign change in late 2021; sentiment and flow ICs near zero; and a 20-day return IC that changed sign in 2022. A liquidity filter weakened carry IC. The draft also reported underperformance against an equal-weight long-only baseline in 2021. These research statistics and baseline comparisons were not independently reproduced in the latest backtest runs.

The submitted strategy returned 31.10% after costs in development, with a Sharpe ratio of 1.112. Validation was nearly flat and deteriorated under higher costs, limiting support for a persistent profitable relationship.

3. Construction

At each close, I rank the four assets in each of six sectors by carry score. I buy the highest-ranked asset and short the lowest-ranked asset for execution at the next open. Asset ID breaks ties; the other two assets have zero target weights.

Base weights are +10% and -10% per pair: 12 positions, 120% gross exposure, and zero sector net exposure before volatility scaling. The 10% base leaves room below the 20% per-asset cap. Position magnitudes are:

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10% x clip(0.00623 / market_volatility, 0.5, 2.0)
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Market volatility is the sample standard deviation of the last 20 market returns, including the current day, with a floor of 0.001. The fixed 0.00623 reference was identified in the research draft as the 2021 median; it was not refitted for these runs.

4. Risk

Six sector pairs spread exposure across 12 assets. Equal long and short weights reduce shared market exposure but do not eliminate differences in asset risk. Requested gross exposure can reach 240%; the engine scales weights to meet its 200% cap. It adjusted exposure on 32 development and 9 validation sessions. Executed net exposure was effectively zero.

The strategy can lose if the carry relationship weakens, selected assets move against the positions, or costs exceed the spread. Maximum drawdown was 6.74% in development and 5.66% in validation at standard costs.

5. Costs

The draft rejected the frequently changing 1-day return ranking partly because of turnover concerns. Carry rankings were more persistent, but rebalancing, volatility scaling and asset selection still generated average daily turnover of 20.30% in development and 19.10% in validation.

At standard costs, summed daily cost fractions were 10.94% in development and 1.51% in validation. Validation costs rose to 2.27% at 1.5x and 3.02% at 2x, while net returns worsened. These costs are sums relative to preceding closing NAV, not differences between compounded gross and net returns. The model includes spread and impact, but excludes financing and stock borrow.

6. Validation

Development scored January 4, 2021-June 28, 2024 (910 sessions); January 1 supplied the initial signal history. Validation scored July 1-December 31, 2024 (132 sessions), with development observations available as history. Each run started from \$1 million in cash and included terminal liquidation costs.

Run	Net return	Sharpe	Max drawdown
Development, 1x	31.10%	1.112	-6.74%
Validation, 1x	-0.08%	0.011	-5.66%
Validation, 1.5x	-0.83%	-0.203	-6.14%
Validation, 2x	-1.58%	-0.418	-6.61%

Returns cover each full period; Sharpe uses 252 sessions per year. All four runs used the same submitted strategy and parameters, with no tuning after inspecting validation in this work. The 1.5x and 2x runs are cost sensitivity checks. Validation did not sustain development performance; it has now been inspected and should not be reused as an untouched test.

7. Limitations

Choosing carry after comparing signals risks overfitting. The validation sample is short, and its weak result limits confidence in persistence. Existing holdings earn overnight returns, so portfolio performance does not isolate the open-to-close hypothesis. Stable rankings do not ensure stable profits, and less liquid assets may be costly to trade. Prior IC, liquidity-filter and baseline claims still need reproducible checks.

8. Next step

Decompose the validation result into overnight and intraday returns for the long and short positions, with trading costs shown separately. This would help determine whether the carry spread weakened or whether overnight exposure and costs offset an intraday advantage.